

DID, Event Studies, and Synthetic Control

Empirical Methods — Week 3

Teaching deck draft

- Which untreated path stands in for treated units' missing untreated outcomes?
- When do panel comparisons become credible causal designs?
- When do fixed effects hide bad comparisons, contaminated dynamics, or fragile inference?

DID, event studies, and synthetic control are counterfactual-path designs, not just panel regressions.

2x2 DID and parallel trends

$$\hat{\tau}^{DID} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]$$

- Identifies a post-period ATT under parallel trends and no anticipation.
- Best for one sharp treatment date with a credible comparison group.
- Does not identify long-run, spillover, equilibrium, or out-of-support effects by itself.

Card–Krueger as design anchor

- Treated path: New Jersey fast-food restaurants after the minimum-wage increase.
- Comparison path: Pennsylvania restaurants over the same window.
- Design claim: Pennsylvania's employment change proxies for New Jersey's missing untreated change.
- Interpretation: the second difference is an ATT for this policy, period, and comparison.

The paper is useful because the counterfactual path is visible enough to debate.

Event studies and dynamic paths

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - G_i = k\} + \varepsilon_{it}$$

- Event time $k = t - G_i$ organizes leads and lags around first treatment.
- Post coefficients describe dynamic response only if comparisons are valid.
- Pre coefficients are diagnostics; insignificant leads do not prove parallel trends.
- Report omitted period, binning, anticipation window, and cohort support.

Why staggered TWFE can fail

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}$$

- TWFE with staggered adoption is a weighted mix of 2x2 comparisons.
- Some comparisons use already-treated units as controls for newly treated units.
- With heterogeneous effects, already-treated outcomes contain treatment effects.
- Event-study coefficients can be contaminated by other cohorts' effects.
- A precise coefficient can answer a comparison the paper does not want.

Modern DID estimator choices

Tool	Best use	What it fixes
Goodman–Bacon	Diagnose static TWFE	Reveals weighted 2x2 comparison structure
Callaway–Sant’Anna	Group-time effects and transparent aggregation	Uses valid not-yet-treated or never-treated controls
Sun–Abraham	Dynamic event-study path	Avoids contamination of relative-time coefficients
Borusyak–Jaravel–Spiess	Imputation-style event studies and aggregates	Reconstructs untreated outcomes from untreated observations

Choose the estimator after naming the estimand and comparison set.

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g], \quad t \geq g$$

$$\widehat{ATT}_S = \frac{1}{|S|} \sum_{(i,t) \in S} (Y_{it} - \hat{Y}_{it}(0))$$

- Group-time DID estimates effects for a cohort at a calendar time.
- Aggregation weights should match the research question.
- Imputation makes the untreated-outcome model explicit.

$$\hat{V}_{CR} = (X'X)^{-1} \left(\sum_{g=1}^G X'_g \hat{u}_g \hat{u}'_g X_g \right) (X'X)^{-1}$$

- DID outcomes and treatments are usually persistent.
- Cluster at the treatment-assignment or residual-dependence level.
- Few clusters or few treated clusters require caution.
- Use wild cluster bootstrap or randomization/placebo inference when appropriate.
- Standard errors quantify uncertainty around an estimand; they do not validate the comparison.

Synthetic control

$$\hat{w} = \arg \min_w (X_1 - X_0 w)' V (X_1 - X_0 w) \quad \text{s.t. } w_j \geq 0, \sum_j w_j = 1$$

$$\hat{\tau}_{1t} = Y_{1t} - \sum_j \hat{w}_j Y_{jt}$$

- Best for one or a few treated aggregate units.
- Pre-treatment fit is part of the design argument.
- Report donor pool, weights, pre-fit, post gaps, and placebo comparisons.

Design checklist

- ① What changed, when, and for whom?
- ② What is the estimand: 2x2 ATT, $ATT(g, t)$, event-time path, or synthetic gap?
- ③ Which units are valid controls at each time?
- ④ Is treatment absorbing, staggered, reversible, or continuous?
- ⑤ Is no anticipation plausible?
- ⑥ What is the clustering or placebo-inference level?
- ⑦ What does the estimate not identify?

Reproduce

- Card–Krueger-style 2x2 DID
- State paths
- Regression equivalence

Diagnose

- Staggered adoption panel
- TWFE vs group-time ATT
- Imputation-style ATT
- Event-time support

Transfer

- Single treated city
- Donor weights
- Pre-fit and post gaps
- Placebo comparisons

Takeaways

- The design object is the missing untreated path for treated units.
- Classical DID is powerful when the comparison trend is credible.
- Staggered TWFE can hide forbidden comparisons and contaminated dynamics.
- Modern DID methods define valid comparisons and transparent aggregations.
- Inference discipline is central when treatment and outcomes are persistent.
- Synthetic control is the same counterfactual-path logic with donor construction.